

HOSPITAL MANAGEMENT ERP

Comprehensive System Documentation

Version 1.0

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1. Executive Summary

This document provides a comprehensive, technology-agnostic specification for a Hospital Management Enterprise Resource Planning (ERP) system. The system is designed to digitize, streamline, and integrate all clinical, administrative, and financial operations of a modern hospital or healthcare facility.

The Hospital Management ERP (hereafter referred to as HM-ERP) covers the complete lifecycle of patient care from registration through discharge and follow-up, while simultaneously managing the business, financial, compliance, and operational aspects of hospital administration.

1.1 Purpose

The purpose of this document is to serve as a definitive reference for all stakeholders involved in the design, development, deployment, and maintenance of the HM-ERP system. It is written to be agnostic of any specific technology stack, allowing implementation teams to choose the most appropriate technologies for their context.

1.2 Scope

The HM-ERP system encompasses the following functional domains:

- Patient Management (Registration, OPD, IPD, Emergency)
- Clinical Operations (Doctor Workflows, Nursing, OT Management)
- Diagnostic Services (Laboratory, Radiology, Pathology)
- Pharmacy and Medication Management
- Billing, Revenue Cycle, and Insurance Management
- Human Resources and Payroll
- Inventory and Supply Chain Management
- Blood Bank Management
- Dietary and Kitchen Management
- Housekeeping and Facility Management
- Quality and Compliance Management
- Reporting, Analytics, and Business Intelligence
- System Administration and Security

1.3 Target Audience

Audience	Usage
Hospital Administration / CXOs	Strategic overview, ROI justification, compliance understanding
Project Managers	Scope definition, module planning, resource estimation
Business Analysts	Requirements validation, workflow mapping, gap analysis

Software Architects	System design, integration patterns, data model design
Development Teams	Feature implementation, API contracts, data schemas
QA / Testing Teams	Test case derivation, acceptance criteria, edge cases
IT Operations / DevOps	Deployment planning, infrastructure sizing, monitoring
Compliance Officers	Regulatory requirements, audit trails, data governance

2. System Overview and Architecture

2.1 Design Principles

The HM-ERP system is built on the following foundational principles:

Modularity: Each functional domain is an independent module with well-defined interfaces. Modules can be deployed, scaled, and updated independently.

Interoperability: The system supports healthcare standards including HL7 FHIR, DICOM, ICD-10/11, SNOMED CT, and LOINC for seamless data exchange with external systems.

Scalability: Architecture supports horizontal scaling from a 50-bed clinic to a 5000+ bed multi-campus hospital network.

Security by Design: Role-based access control (RBAC), data encryption at rest and in transit, comprehensive audit logging, and compliance with HIPAA, GDPR, and local healthcare regulations.

High Availability: Designed for 99.9%+ uptime with failover mechanisms, data replication, and disaster recovery capabilities.

Offline Capability: Critical modules (Emergency, Pharmacy) support offline operation with automatic sync when connectivity is restored.

Multi-tenancy: Supports multiple hospitals, branches, and departments within a single deployment.

2.2 High-Level Architecture

The system follows a layered architecture pattern with clear separation of concerns:

Layer	Responsibility	Key Components
Presentation Layer	User interfaces for all device types	Web Portal, Mobile Apps (iOS/Android), Kiosk UI, Tablet Interface
API Gateway Layer	Request routing, auth, rate limiting	API Gateway, Load Balancer, Auth Service
Application Layer	Business logic and workflow orchestration	Module Services, Workflow Engine, Notification Service, Scheduling Engine
Integration Layer	External system communication	HL7/FHIR Adapter, PACS Integration, Payment Gateway, Lab Instrument Interface
Data Layer	Persistence and caching	Primary Database, Read Replicas, Cache, Document Store, Search Index
Infrastructure Layer	Runtime environment and operations	Container Orchestration, Message Queue, Object Storage, CDN, Monitoring

2.3 Module Dependency Map

The following table shows inter-module dependencies. Understanding these is critical for phased implementation planning.

Module	Depends On	Depended By
Patient Registration	Admin, Master Data	All clinical and billing modules
OPD	Patient Registration, Doctor Master	Pharmacy, Lab, Radiology, Billing
IPD	Patient Registration, Bed Management	Nursing, OT, Dietary, Billing
Laboratory	Patient Registration, OPD/IPD	Billing, Reports
Radiology	Patient Registration, OPD/IPD	Billing, Reports
Pharmacy	Inventory, Patient Registration	Billing, IPD
Billing	All clinical modules	Insurance, Accounts, Reports
Inventory	Purchase, Vendor Master	Pharmacy, OT, Lab
HR & Payroll	Admin	Accounts, Reports
Blood Bank	Patient Registration, Lab	OT, IPD, Billing
Insurance / TPA	Billing, Patient Registration	Accounts, Reports
Admin & Security	None (foundational)	All modules

2.4 Integration Standards

The HM-ERP system integrates with external systems using the following standards and protocols:

Standard / Protocol	Purpose	Usage Context
HL7 FHIR R4	Healthcare data exchange	Patient records, clinical data, interoperability with other HIS
DICOM 3.0	Medical imaging	Radiology integration with PACS, image viewing
ICD-10 / ICD-11	Disease classification	Diagnosis coding, insurance claims, reporting
SNOMED CT	Clinical terminology	Clinical documentation, decision support
LOINC	Lab observation coding	Laboratory test catalogs, result coding
CPT / HCPCS	Procedure coding	Billing, insurance claims
ABDM (India)	National health ID	Ayushman Bharat Digital Mission compliance
NABL / CAP	Lab accreditation	Quality management, lab workflows
NABH	Hospital accreditation	Quality indicators, patient safety protocols

ISO 27001	Information security	Security controls, audit requirements
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3. User Roles and Access Control

3.1 Role Hierarchy

The system implements a hierarchical RBAC model. Each role has a predefined set of permissions that can be customized per deployment.

Role Category	Roles	Primary Access
Super Admin	System Administrator	Full system configuration, user management, audit logs, all modules
Hospital Admin	Hospital Director, COO, CMO	Dashboard, reports, approvals, policy configuration
Clinical - Doctors	Consultant, Resident, Surgeon, Anesthetist	OPD/IPD, prescriptions, clinical notes, OT scheduling
Clinical - Nursing	Head Nurse, Staff Nurse, Ward In-Charge	Patient vitals, medication administration, care plans, bed management
Diagnostics	Lab Technician, Radiologist, Pathologist	Sample collection, test results, image interpretation
Pharmacy	Chief Pharmacist, Pharmacist, Pharmacy Technician	Drug dispensing, inventory, prescription validation
Front Office	Receptionist, Registration Clerk, Appointment Coordinator	Patient registration, appointments, visitor management
Billing	Billing Executive, Revenue Manager, Insurance Coordinator	Invoice generation, payments, insurance claims, refunds
HR & Admin	HR Manager, Payroll Admin, Department Head	Employee management, attendance, payroll, leave management
Inventory	Store Manager, Purchase Officer, GRN Clerk	Purchase orders, stock management, vendor management
Blood Bank	Blood Bank Officer, Technician	Donor management, cross-matching, issue/return
Dietary	Dietician, Kitchen Manager	Diet plans, meal ordering, kitchen management
Quality	Quality Manager, Infection Control Nurse	Quality indicators, incident reporting, compliance tracking
Patient / Visitor	Patient, Attendant	Patient portal: appointments, reports, bills, feedback

3.2 Permission Model

Permissions are defined at four levels: Module access, Feature access within a module, Action permissions (Create, Read, Update, Delete, Print, Export), and Data scope (Own records, Department, Branch, Organization-wide).

3.3 Authentication Requirements

- Multi-factor authentication (MFA) for all administrative and clinical roles
- Single Sign-On (SSO) support via SAML 2.0 and OpenID Connect
- Session timeout policies configurable per role (default: 15 min for clinical, 30 min for admin)
- Biometric authentication support for high-security areas (pharmacy narcotics, blood bank)
- Emergency break-glass access with mandatory audit logging
- IP-based and device-based access restrictions
- Password complexity policies with rotation enforcement

4. Patient Registration Module

4.1 Overview

The Patient Registration module serves as the entry point for all patient interactions with the hospital. It maintains a master patient index (MPI) and assigns a unique hospital registration number (UHID) to each patient. This module is foundational; all clinical and financial modules depend on patient data.

4.2 Functional Requirements

4.2.1 New Patient Registration

- Capture patient demographics: name, date of birth, gender, blood group, marital status, nationality
- Contact information: phone numbers, email, emergency contact details
- Address: current and permanent address with geocoding support
- Identity documents: national ID, passport, insurance card, driving license (with document scan)
- Photo capture via webcam or upload
- Next of kin / guardian information (mandatory for minors)
- Referral source tracking: self, doctor referral, hospital referral, insurance, corporate
- Allergy and chronic condition quick-capture at registration
- Consent capture: treatment consent, data sharing consent, research participation
- Auto-generation of UHID with configurable format (e.g., HOSP-2026-000001)

4.2.2 Patient Search and Retrieval

- Search by UHID, name, phone number, national ID, date of birth
- Fuzzy search / phonetic matching for name searches
- Duplicate detection with merge capability
- Recent patients quick-access list
- Barcode and QR code scanning for patient identification

4.2.3 Visit Management

- Track visit type: OPD, IPD, Emergency, Day Care, Health Checkup
- Visit history with timeline view
- Re-registration for returning patients (pre-populated data)
- Multi-visit support (patient can have concurrent OPD and IPD visits)

4.3 Data Model

Entity	Key Attributes	Relationships
Patient	UHID, name, DOB, gender, blood_group, photo, status	Has many Visits, Contacts, Documents, Allergies
PatientContact	type, phone, email, is_primary, is_emergency	Belongs to Patient
PatientAddress	type, line1, line2, city, state, country, pin_code, geo_lat, geo_lng	Belongs to Patient
PatientDocument	type, document_number, issuing_authority, expiry, scan_file	Belongs to Patient
PatientAllergy	allergen, type (drug/food/env), severity, reaction, verified_by	Belongs to Patient
Visit	visit_id, type, department, doctor, status, admitted_at, discharged_at	Belongs to Patient, has many Orders
NextOfKin	name, relationship, phone, address, is_guardian	Belongs to Patient
Consent	type, version, status, signed_at, witness, document_file	Belongs to Patient, Visit

4.4 Workflows

4.4.1 New Patient Registration Flow

1. Front desk verifies if patient exists (search by name, phone, ID)
2. If new: collect demographics, photo, documents, contacts
3. System checks for potential duplicates and alerts operator
4. Generate UHID and print patient card with barcode/QR
5. Capture initial consents
6. Create visit record and route to appropriate department (OPD/ER/IPD)
7. Notify assigned department of new patient arrival

4.4.2 Duplicate Detection and Merge Flow

8. System flags potential duplicates based on name + DOB + phone matching
9. Authorized user reviews flagged records side-by-side
10. Select master record and merge secondary records
11. All visit history, billing, and clinical data consolidated under master UHID
12. Audit trail created for the merge operation
13. Secondary UHIDs marked as inactive with redirect pointer

4.5 API Specifications

Endpoint	Method	Description	Auth Required
/api/v1/patients	POST	Register new patient	Registration Clerk+
/api/v1/patients	GET	Search patients with filters	Any authenticated user
/api/v1/patients/{uhid}	GET	Get patient details	Clinical/Admin role
/api/v1/patients/{uhid}	PUT	Update patient information	Registration Clerk+
/api/v1/patients/{uhid}/visits	POST	Create new visit	Registration Clerk+
/api/v1/patients/{uhid}/visits	GET	Get visit history	Clinical/Admin role
/api/v1/patients/{uhid}/allergies	GET/ POST	Manage allergies	Clinical role
/api/v1/patients/{uhid}/consents	POST	Record consent	Registration Clerk+
/api/v1/patients/duplicates	GET	Get potential duplicate list	Admin role
/api/v1/patients/merge	POST	Merge duplicate records	Admin role

5. Appointment and Scheduling Module

5.1 Overview

Manages scheduling of patient appointments, doctor availability, resource allocation (rooms, equipment), and handles the complete appointment lifecycle from booking through completion.

5.2 Functional Requirements

- Doctor schedule management: weekly templates, overrides, leave integration
- Multi-slot types: regular consultation, follow-up (shorter), procedure, teleconsultation
- Online booking via patient portal and mobile app
- Walk-in token generation with estimated wait time
- Overbooking controls with configurable limits per doctor
- Recurring appointment support (e.g., dialysis 3x/week)
- Appointment reminders via SMS, email, WhatsApp, push notification
- Cancellation and rescheduling with reason tracking
- No-show tracking and flagging
- Doctor-to-doctor referral scheduling
- Resource scheduling: rooms, equipment, OT slots
- Multi-doctor appointments (e.g., tumor board)
- Queue management with digital display integration

5.3 Data Model

Entity	Key Attributes	Relationships
DoctorSchedule	doctor_id, day_of_week, start_time, end_time, slot_duration, max_patients, location	Belongs to Doctor, Department
ScheduleOverride	doctor_id, date, type (holiday/extra/modified), override_slots	Belongs to DoctorSchedule
Appointment	appointment_id, patient_id, doctor_id, slot_time, type, status, source	Belongs to Patient, Doctor
Token	token_number, appointment_id, counter, status, called_at, seen_at	Belongs to Appointment
WaitingQueue	department_id, current_token, avg_wait_time, display_config	Has many Tokens

5.4 API Specifications

Endpoint	Method	Description
/api/v1/appointments	POST	Book new appointment

/api/v1/appointments/{id}	PUT	Reschedule or update appointment
/api/v1/appointments/{id}/cancel	POST	Cancel appointment with reason
/api/v1/appointments/{id}/checkin	POST	Patient check-in, generate token
/api/v1/doctors/{id}/availability	GET	Get available slots for a date range
/api/v1/doctors/{id}/schedule	GET/ PUT	Manage doctor schedule templates
/api/v1/queue/{department_id}	GET	Get current queue status and wait times

6. Outpatient Department (OPD) Module

6.1 Overview

The OPD module manages the complete outpatient workflow from patient arrival through consultation, investigation ordering, prescription, and follow-up scheduling. It serves as the primary clinical workflow for doctors and is the highest-volume module in the system.

6.2 Functional Requirements

6.2.1 Doctor Workstation

- Patient queue dashboard with color-coded priority indicators
- Quick access to patient history: past visits, investigations, medications, allergies
- Clinical documentation with customizable templates per specialty
- Chief complaints capture with ICD-10/SNOMED coding
- Vitals display (integrated from nursing assessment)
- Examination findings with body diagram annotation
- Diagnosis entry with ICD-10 coding (primary and secondary diagnoses)
- Investigation ordering: lab, radiology, pathology with urgency flags
- Prescription writing with drug interaction and allergy checks
- Procedure documentation
- Referral to other departments or specialists
- Follow-up scheduling
- Medical certificate and fitness certificate generation
- Clinical notes: SOAP format, free text, voice dictation support

6.2.2 Prescription Management

- Drug search with generic name, brand name, and formulary flag
- Dosage templates: standard dosing based on indication
- Drug-drug interaction checking (real-time)
- Drug-allergy cross-referencing
- Renal/hepatic dose adjustment alerts
- Pregnancy and lactation safety warnings
- Favorite prescription templates per doctor
- Previous prescription copy and modify
- E-prescription with pharmacy routing

6.3 Data Model

Entity	Key Attributes
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OPDVisit	visit_id, patient_id, doctor_id, department, chief_complaint, vitals_snapshot, status, notes
ClinicalNote	note_id, visit_id, type (SOAP/free), subjective, objective, assessment, plan, created_by
Diagnosis	visit_id, icd_code, description, type (primary/secondary/differential), certainty
Prescription	prescription_id, visit_id, doctor_id, status, pharmacy_status, notes
PrescriptionItem	drug_id, dose, frequency, route, duration, quantity, instructions, substitution_allowed
InvestigationOrder	order_id, visit_id, test_id, urgency, clinical_info, status, results_id
Referral	from_doctor, to_doctor/dept, reason, urgency, clinical_summary, status

6.4 Clinical Decision Support

The OPD module integrates clinical decision support (CDS) features to enhance patient safety:

- Real-time drug interaction alerts (severity: minor, moderate, major, contraindicated)
- Allergy cross-check before any drug or investigation order
- Duplicate order detection (same test ordered within configurable window)
- Critical value notification for lab results
- Evidence-based clinical guidelines integration
- Antimicrobial stewardship alerts

7. Inpatient Department (IPD) Module

7.1 Overview

The IPD module manages the entire inpatient journey from admission through treatment, monitoring, and discharge. It coordinates activities across nursing, dietary, pharmacy, and diagnostic departments.

7.2 Functional Requirements

7.2.1 Admission

- Admission types: planned, emergency, transfer-in, day care
- Bed selection with real-time availability: ward, semi-private, private, ICU, NICU, PICU
- Admitting doctor and primary consultant assignment
- Advance deposit collection
- Admission orders: diet, nursing care plan, standing medication orders
- Medico-legal case (MLC) flagging with police intimation workflow
- Estimated treatment cost and duration

7.2.2 Bed Management

- Visual floor map with real-time bed status (available, occupied, reserved, maintenance, housekeeping)
- Bed categories: general ward, semi-private, private, deluxe, suite, ICU, isolation
- Bed transfer with reason tracking and automatic billing rate adjustment
- Bed reservation for planned admissions and post-OT
- Bed turnaround tracking: discharge to clean to available
- Occupancy analytics by ward, floor, category

7.2.3 Doctor Rounds / Clinical Management

- Daily round notes with assessment and plan updates
- Progress notes with trending of vitals and lab values
- Consultant-to-consultant referral within admission
- Multi-disciplinary team notes
- Procedure/surgery planning and consent
- Discharge planning from day of admission

7.2.4 Nursing Workflows

- Initial nursing assessment with standardized assessment tools
- Vital signs monitoring with configurable frequency per patient acuity

- Intake-output charting
- Medication administration record (MAR) with barcode scanning
- Nursing care plans and interventions
- Pain assessment (numeric, FLACC, Wong-Baker scales)
- Fall risk assessment (Morse scale)
- Pressure ulcer risk assessment (Braden scale)
- Shift handover with structured handoff (SBAR format)
- Nursing shift assignment and patient allocation

7.2.5 Discharge

- Discharge types: normal, against medical advice (AMA), death, transfer-out, absconded
- Discharge summary generation with configurable template
- Final billing settlement
- Discharge medication list and instructions
- Follow-up appointment scheduling
- Patient education materials
- Feedback collection
- Death certificate and cause of death coding (ICD-10)

7.3 Data Model

Entity	Key Attributes
Admission	admission_id, patient_id, type, admitting_doctor, primary_consultant, bed_id, status, admitted_at, discharged_at
Bed	bed_id, ward_id, floor, room_number, bed_number, category, status, features (oxygen, suction)
Ward	ward_id, name, floor, department, total_beds, nurse_station
BedTransfer	from_bed, to_bed, reason, transferred_at, authorized_by
NursingAssessment	assessment_id, admission_id, type, score, findings, assessed_by, assessed_at
VitalSign	admission_id, datetime, temperature, pulse, bp_systolic, bp_diastolic, spo2, respiratory_rate, recorded_by
IntakeOutput	admission_id, datetime, type (intake/output), category, volume_ml, route, recorded_by
MedicationAdministration	prescription_item_id, admission_id, scheduled_time, given_time, status, given_by, witnessed_by
DischargeSummary	admission_id, summary_text, diagnosis_codes, procedures, discharge_meds, instructions, prepared_by

8. Emergency Department Module

8.1 Overview

The Emergency module handles high-acuity, time-sensitive patient care. It provides rapid triage, tracking, and disposition workflows optimized for the chaotic environment of an emergency department.

8.2 Functional Requirements

- Rapid registration (minimal data set: name, age, gender, chief complaint, contact)
- Triage using standardized systems: ESI (Emergency Severity Index) 1-5, or Manchester Triage
- Color-coded patient tracking board (critical, urgent, semi-urgent, non-urgent)
- Timer-based alerts for patients exceeding triage-defined wait times
- Trauma activation protocols with team notification
- Resuscitation documentation with timestamped interventions
- MLC (Medico-Legal Case) registration with mandatory fields
- Disposition tracking: admit, discharge, transfer, LAMA, death
- Ambulance tracking and pre-arrival notification
- Mass casualty incident (MCI) mode with simplified workflows
- Integration with bed management for emergency admissions
- Poison control reference database
- Emergency drug dosing calculator (weight-based)

8.3 Triage Classification

ESI Level	Category	Target Response Time	Examples
1	Resuscitation	Immediate	Cardiac arrest, major trauma, severe respiratory distress
2	Emergent	< 10 minutes	Chest pain, stroke symptoms, severe allergic reaction
3	Urgent	< 30 minutes	Abdominal pain, moderate trauma, high fever
4	Less Urgent	< 60 minutes	Minor fractures, mild infections, simple lacerations
5	Non-Urgent	< 120 minutes	Minor complaints, medication refills, routine follow-up

9. Operation Theatre (OT) Management Module

9.1 Overview

Manages the surgical workflow from scheduling through post-operative care, including OT resource management, surgical team coordination, anesthesia documentation, and safety checklists.

9.2 Functional Requirements

- OT scheduling with real-time availability calendar
- Surgery request and approval workflow
- Pre-operative assessment checklist (WHO Surgical Safety Checklist)
- Surgical team composition: surgeon, assistant, anesthetist, scrub nurse, circulating nurse
- Anesthesia pre-assessment and documentation
- OT equipment and implant tracking
- Intra-operative documentation: procedure notes, anesthesia record, blood products used
- Surgical specimen tracking to pathology
- Sponge and instrument count verification
- Post-operative orders and recovery monitoring (PACU)
- OT utilization analytics: on-time starts, turnaround time, cancellation rates
- Emergency surgery priority override
- Surgeon preference cards (standing orders for equipment, sutures, implants)

9.3 Surgical Safety Checklist

Phase	Checkpoints
Sign In (Before Anesthesia)	Patient identity confirmed, site marked, consent signed, pulse oximeter on, allergy check, airway assessment, blood loss risk assessed
Time Out (Before Incision)	Team introduction, patient identity re-confirmed, procedure and site verified, antibiotic prophylaxis given, critical steps reviewed, imaging displayed
Sign Out (Before Patient Leaves OT)	Procedure name recorded, instrument/sponge/needle count correct, specimen labeled, equipment issues noted, post-op orders reviewed

10. Laboratory Information System (LIS) Module

10.1 Overview

The Laboratory module manages the complete lifecycle of diagnostic tests from order entry through sample collection, processing, result entry, validation, and reporting. It supports multiple lab sections: biochemistry, hematology, microbiology, serology, histopathology, and cytology.

10.2 Functional Requirements

10.2.1 Test Catalog Management

- Test master with LOINC coding
- Panel/profile definitions (e.g., Liver Function Test = 8 individual parameters)
- Reference ranges by age, gender, and special conditions (pregnancy)
- Critical value definitions per parameter
- Sample type requirements (serum, plasma, whole blood, urine, CSF, etc.)
- Container type and volume requirements
- Special instructions (fasting, timing, transport conditions)
- Turnaround time (TAT) targets per test
- Test pricing with multiple price lists (cash, insurance, package)

10.2.2 Sample Lifecycle

- Sample collection with barcode label generation
- Phlebotomy queue management
- Sample rejection with reason documentation (hemolyzed, insufficient, wrong container)
- Sample tracking through all phases: collected, received at lab, processing, completed
- Aliquot management for samples requiring multiple sections
- Sample storage location tracking
- External lab send-out management

10.2.3 Result Management

- Manual result entry with normal range highlighting
- Instrument interface for auto-population of results (uni/bi-directional)
- Delta check: comparison with patient previous values
- Auto-validation rules for results within acceptable range
- Technical validation by lab technician
- Medical validation by pathologist/senior tech
- Critical value notification workflow (automated alert to ordering doctor and nurse)
- Amendment workflow with original value retention
- Report formatting with configurable templates per section

- Antibiotic sensitivity reporting (for microbiology)
- Histogram and graph attachment (hematology analyzers)

10.3 Data Model

Entity	Key Attributes
TestMaster	test_id, name, loinc_code, section, sample_type, container, volume_ml, tat_hours, price, status
TestParameter	parameter_id, test_id, name, unit, data_type, reference_ranges[], critical_values
TestPanel	panel_id, name, tests[], price
LabOrder	order_id, visit_id, patient_id, ordered_by, tests[], urgency, clinical_info, status
Sample	sample_id, barcode, order_id, type, collected_by, collected_at, received_at, status
LabResult	result_id, order_id, test_id, parameter_values[], tech_validated_by, doctor_validated_by, status
CriticalAlert	result_id, parameter, value, notified_to, acknowledged_at, acknowledged_by
QualityControl	qc_id, instrument, test, level, expected_range, observed_value, status, action_taken

10.4 Instrument Integration

The LIS supports bi-directional interfacing with laboratory instruments:

Direction	Function	Protocol Support
Host to Instrument	Send sample/test orders with barcode	ASTM/LIS2-A2, HL7 v2.x, Serial RS-232
Instrument to Host	Receive results automatically	ASTM/LIS2-A2, HL7 v2.x, TCP/IP
Bi-directional	Order download + result upload	Middleware integration, API-based

11. Radiology Information System (RIS) Module

11.1 Overview

The Radiology module manages imaging workflows including order management, scheduling, image acquisition, reporting, and PACS integration. It covers all imaging modalities: X-ray, CT, MRI, Ultrasound, Mammography, PET-CT, Fluoroscopy, and Interventional Radiology.

11.2 Functional Requirements

- Imaging order management with clinical indication and priority
- Modality worklist (MWL) via DICOM for automated patient/study push to machines
- Scheduling for modalities requiring appointment (MRI, CT, special procedures)
- Patient preparation instructions (contrast, fasting, bowel prep)
- Contrast administration documentation and adverse reaction tracking
- Radiation dose tracking and cumulative dose monitoring
- PACS integration for image storage and retrieval (DICOM 3.0)
- Radiology reporting with structured and narrative formats
- Report templates per modality and body part
- Preliminary read by resident with attending radiologist final verification
- Critical finding notification (stroke, PE, pneumothorax, etc.)
- Prior study comparison linking
- Teleradiology support for remote reading
- Film/CD delivery tracking

11.3 DICOM Integration

Service	Purpose
DICOM Modality Worklist (MWL)	Push patient and order details to imaging equipment
DICOM Storage (C-STORE)	Send acquired images to PACS
DICOM Query/Retrieve (C-FIND/C-MOVE)	Search and retrieve prior studies from PACS
DICOM Print (C-PRINT)	Print images on film printer
DICOM SR (Structured Report)	Structured reporting format for measurements and findings
WADO-RS / DICOMweb	Web-based image access for viewer integration

12. Pharmacy Module

12.1 Overview

The Pharmacy module manages medication dispensing for both outpatients and inpatients, drug inventory control, formulary management, and medication safety systems. It integrates tightly with OPD prescriptions, IPD medication orders, and the inventory module.

12.2 Functional Requirements

12.2.1 Drug Master and Formulary

- Drug catalog with generic name, brand names, manufacturer, formulation, strength
- Drug classification: therapeutic class, pharmacological class, schedule (narcotic, controlled)
- Formulary management with approved/restricted/non-formulary status
- Drug interaction database with severity levels
- Therapeutic equivalence mapping
- Bar code / GTIN mapping for each drug pack
- Look-alike, sound-alike (LASA) drug flagging
- High-alert medication identification

12.2.2 Dispensing

- Outpatient dispensing against OPD prescriptions
- Inpatient dispensing: ward stock, unit dose, and floor stock models
- Barcode verification at dispensing
- Substitution management with pharmacist authorization
- Partial dispensing with balance tracking
- Return to pharmacy with reason documentation
- Narcotic and controlled substance dispensing with double verification
- Discharge medication preparation
- Patient counseling documentation

12.2.3 Inventory Control

- Real-time stock levels across all pharmacy locations (main, satellite, emergency)
- Batch and expiry tracking
- Near-expiry alerts (configurable: 30/60/90 days)
- Auto-reorder point and economic order quantity (EOQ) calculation
- Inter-pharmacy stock transfer
- Drug recall management
- ABC/VED analysis for inventory optimization

- Temperature-sensitive drug monitoring

12.3 Data Model

Entity	Key Attributes
DrugMaster	drug_id, generic_name, brand_name, manufacturer, formulation, strength, unit, schedule, is_formulary
DrugInteraction	drug_a_id, drug_b_id, severity, description, clinical_action
PharmacyStock	drug_id, location_id, batch_number, expiry_date, quantity, purchase_price, mrp
Dispensing	dispensing_id, prescription_id, patient_id, drug_id, quantity, batch, dispensed_by, verified_by
NarcoticRegister	drug_id, date, opening_stock, received, dispensed, closing_stock, verified_by, witness

13. Billing and Revenue Cycle Module

13.1 Overview

The Billing module is the financial backbone of the HM-ERP. It handles charge capture from all service-providing modules, invoice generation, payment collection, refunds, and integration with insurance and accounting systems.

13.2 Functional Requirements

13.2.1 Charge Master and Tariff Management

- Service charge master with CPT/HCPCS coding
- Multiple tariff structures: cash, insurance, corporate, government scheme, package
- Department-wise, doctor-wise, and room-category-wise rate variation
- Package rate definitions (e.g., knee replacement all-inclusive)
- Seasonal and promotional pricing
- Tax configuration (GST, service tax) per service category
- Price list version control with effective dates

13.2.2 Charge Capture

- Automatic charge capture from OPD consultation, lab orders, radiology orders, pharmacy dispensing
- IPD per-day charges: bed, nursing, monitoring
- OT charges: surgeon fee, anesthesia, OT usage, consumables, implants
- Professional fees (doctor fees) management
- Manual charge entry for services not covered by other modules
- Charge bundling and unbundling rules

13.2.3 Invoice and Payment

- OPD invoice generation (real-time at each service point or consolidated)
- IPD interim and final bill generation
- Payment modes: cash, card, UPI, bank transfer, cheque, wallet
- Advance/deposit collection and adjustment
- Split billing: patient share + insurance share + corporate share
- Discount management with approval workflow
- Refund processing with authorization
- Receipt and invoice printing with configurable templates
- Payment gateway integration for online payments

13.2.4 Insurance and TPA Processing

- Pre-authorization request and approval tracking
- Claim generation in standard formats (per insurer requirements)
- Claim submission tracking and follow-up
- Cashless vs reimbursement billing
- Insurance rejection management and re-submission
- Settlement reconciliation
- TPA-wise accounts receivable aging

13.3 Data Model

Entity	Key Attributes
ChargeMaster	service_id, name, cpt_code, department, base_rate, tax_percent, effective_from
Tariff	tariff_id, name, type (cash/insurance/corp), effective_from, status
TariffRate	tariff_id, service_id, rate, discount_percent
Invoice	invoice_id, patient_id, visit_id, type (OPD/IPD), total, tax, discount, net_payable, status
InvoiceLineItem	invoice_id, service_id, quantity, rate, amount, tax, discount, department, doctor_id
Payment	payment_id, invoice_id, amount, mode, reference_number, collected_by, collected_at
Advance	advance_id, patient_id, admission_id, amount, payment_mode, status (active/adjusted/refunded)
InsuranceClaim	claim_id, invoice_id, policy_id, pre_auth_number, claimed_amount, approved_amount, status
Refund	refund_id, invoice_id, amount, reason, approved_by, processed_at, mode

14. Inventory and Supply Chain Module

14.1 Overview

Manages the procurement, storage, distribution, and consumption tracking of all hospital supplies including medical consumables, surgical items, general stores, linen, and fixed assets.

14.2 Functional Requirements

14.2.1 Master Data

- Item master with categorization: medical consumables, surgical, general, linen, stationery, implants
- Vendor master with performance scoring
- Rate contracts and rate comparison
- Store/warehouse locations

14.2.2 Procurement

- Purchase indent from departments
- Purchase order generation with approval workflow
- Comparative quotation analysis
- Goods receipt note (GRN) with quality inspection
- Invoice matching: 3-way (PO, GRN, Invoice)
- Vendor payment tracking
- Purchase return management

14.2.3 Stock Management

- Multi-store inventory with real-time stock visibility
- Batch and expiry tracking for all medical items
- Min-max stock levels with auto-reorder alerts
- Stock transfer between stores and departments
- Stock consumption entry by department
- Physical stock verification and adjustment
- FIFO/FEFO issue policies
- Condemned/expired item write-off workflow

14.2.4 Analytics

- Stock valuation reports (weighted average, FIFO)
- ABC analysis by consumption value
- VED analysis (Vital, Essential, Desirable)

- Slow-moving and non-moving item identification
- Consumption trend analysis
- Vendor performance scorecards

15. Human Resources and Payroll Module

15.1 Overview

Manages the complete employee lifecycle from recruitment through retirement, including attendance, leave management, payroll processing, performance management, and regulatory compliance.

15.2 Functional Requirements

15.2.1 Employee Management

- Employee master: personal details, qualifications, certifications, license numbers
- Department and designation management
- Organization hierarchy and reporting structure
- Document management (offer letter, contracts, certifications, license renewals)
- Employee self-service portal
- Onboarding and offboarding checklists

15.2.2 Attendance and Scheduling

- Integration with biometric devices and access control systems
- Shift management for 24/7 hospital operations
- Roster creation with auto-scheduling based on skill requirements
- Overtime tracking and approval
- On-call duty scheduling
- Locum doctor schedule management

15.2.3 Leave Management

- Leave types: casual, sick, earned, maternity, paternity, comp-off, study leave
- Leave balance tracking with carry-forward rules
- Leave request and approval workflow
- Leave calendar with department-wise view
- Holiday calendar management

15.2.4 Payroll

- Salary structure configuration: basic, allowances, deductions
- Payroll processing with attendance integration
- Tax computation and declaration management
- Statutory compliance: provident fund, professional tax, ESI (India)
- Salary revision and increment management

- Full and final settlement processing
- Pay slip generation and distribution
- Bank file generation for salary disbursement

15.2.5 Credentialing

- Doctor and nurse license verification and tracking
- License renewal reminders
- Continuing Medical Education (CME) credit tracking
- Privilege delineation for clinical staff
- Background verification tracking

16. Blood Bank Module

16.1 Overview

Manages the complete blood banking workflow including donor management, blood collection, testing, component preparation, cross-matching, issue, and transfusion tracking. Ensures compliance with blood banking regulations and safety standards.

16.2 Functional Requirements

- Donor registration and screening (questionnaire, hemoglobin check, vitals)
- Donor deferral management (temporary and permanent) with reason tracking
- Blood collection with bag barcode tracking
- Blood grouping and typing (ABO, Rh, antibody screening)
- Infectious disease testing (HIV, HBV, HCV, syphilis, malaria)
- Component preparation: whole blood, packed RBC, FFP, platelets, cryoprecipitate
- Blood component inventory with expiry tracking
- Cross-match request processing
- Blood issue with compatibility verification
- Transfusion reaction reporting and investigation
- Blood return and re-issue workflow
- Donor camp management
- External blood bank requisition and receipt
- Blood wastage tracking and analysis
- FIFO-based issue for optimal utilization
- Lookback and traceback investigations

16.3 Data Model

Entity	Key Attributes
Donor	donor_id, name, blood_group, DOB, gender, last_donation_date, deferral_status
DonorScreening	screening_id, donor_id, date, hemoglobin, bp, pulse, questionnaire_status, eligible
BloodBag	bag_id, barcode, donor_id, collection_date, blood_group, volume_ml, status
BloodComponent	component_id, bag_id, type, prepared_date, expiry_date, volume, status, storage_location
TTIResult	bag_id, test, result, method, tested_by, tested_at
CrossMatch	request_id, patient_id, component_id, compatibility, method, tested_by, valid_until
BloodIssue	issue_id, component_id, patient_id, admission_id, issued_to_unit,

	issued_by
TransfusionReaction	reaction_id, issue_id, type, severity, description, action_taken

17. Dietary and Kitchen Management Module

17.1 Overview

Manages diet planning for inpatients, kitchen operations, meal preparation tracking, and special dietary requirements. Integrates with the IPD module for patient diet orders.

17.2 Functional Requirements

- Diet type master: regular, diabetic, renal, cardiac, liquid, soft, NPO (nil per oral)
- Doctor-ordered diet with specific restrictions and allergies
- Dietician assessment and calorie/nutrition planning
- Meal timing configuration per ward/floor
- Kitchen preparation list generation by meal time
- Special meal requests and modifications
- Meal delivery tracking to ward/bed level
- Patient food allergy integration
- Nutrition analysis and calorie tracking
- Kitchen inventory management
- Meal satisfaction survey

18. Quality and Compliance Module

18.1 Overview

Tracks quality indicators, manages incident reporting, supports accreditation requirements (NABH, JCI), and ensures regulatory compliance across all hospital operations.

18.2 Functional Requirements

18.2.1 Quality Indicators

- Clinical quality indicators: hospital-acquired infection rate, surgical site infection rate, mortality rate, readmission rate
- Patient safety indicators: falls, medication errors, near-misses, adverse events
- Operational indicators: average length of stay, OT utilization, bed occupancy rate
- Patient experience: satisfaction scores, complaint resolution time
- Configurable indicator definitions with target thresholds
- Automated data collection from clinical modules
- Trend analysis with drill-down capability

18.2.2 Incident Reporting

- Anonymous and identified incident reporting
- Incident classification: medication error, fall, surgical complication, equipment failure, etc.
- Root cause analysis (RCA) workflow
- Corrective and preventive action (CAPA) tracking
- Incident severity grading
- Investigation assignment and timeline tracking

18.2.3 Infection Control

- Surveillance of hospital-acquired infections (HAI)
- Antibigram generation and tracking
- Hand hygiene compliance monitoring
- Isolation precaution management
- Outbreak detection and management

18.2.4 Accreditation Support

- NABH (National Accreditation Board for Hospitals) standard mapping
- JCI (Joint Commission International) standard mapping
- Document control management (policies, SOPs, work instructions)

- Internal audit scheduling and tracking
- Non-conformity management
- Staff training compliance tracking
- Credential verification audit trails

19. Reporting and Analytics Module

19.1 Overview

Provides comprehensive reporting, dashboards, and business intelligence capabilities across all modules. Supports operational, clinical, financial, and regulatory reporting requirements.

19.2 Report Categories

Category	Key Reports
Clinical	Department-wise patient statistics, diagnosis frequency, procedure volumes, length of stay analysis, mortality review, readmission analysis
Financial	Revenue by department/doctor/service, outstanding receivables, collection efficiency, insurance claim status, cost-per-patient analysis
Operational	Bed occupancy, OT utilization, appointment no-show rates, average wait times, turnaround times (lab, radiology), discharge TAT
Inventory	Stock valuation, consumption analysis, expiry alerts, purchase analysis, vendor performance
HR	Headcount, attrition, leave utilization, overtime analysis, training compliance
Regulatory	Birth and death registry, notifiable disease reporting, MLC register, blood bank statistics, narcotic consumption
Quality	Infection rates, incident summary, patient satisfaction trends, accreditation compliance scores
Executive	KPI dashboard, revenue trends, patient volume trends, department comparison, benchmark analysis

19.3 Dashboard Requirements

- Role-based dashboards: CEO/CMO dashboard, department head dashboard, doctor dashboard, nurse dashboard
- Real-time operational dashboard: current census, ER wait times, bed availability, OT status
- Financial dashboard: daily revenue, collections, outstanding, cash flow
- Configurable widgets with drag-and-drop layout
- Drill-down from summary to detail
- Auto-refresh with configurable intervals
- Export to PDF, Excel, and CSV
- Scheduled report delivery via email
- Ad-hoc report builder for power users

20. System Administration Module

20.1 Overview

Provides centralized system configuration, user management, master data management, and operational controls for the entire HM-ERP platform.

20.2 Functional Requirements

20.2.1 Organization Setup

- Hospital profile: name, address, logo, license numbers, registration details
- Multi-branch/campus configuration
- Department and sub-department setup
- Location hierarchy: campus, building, floor, wing, ward, room, bed
- Fiscal year and financial period configuration

20.2.2 Master Data Management

- ICD-10/11 code master with search
- Drug master management
- Investigation master (lab tests, radiology examinations)
- Service and procedure master
- Vendor master
- Insurance company and TPA master
- Corporate and employer master
- Referring doctor master

20.2.3 User Management

- User creation with role assignment
- Role and permission configuration
- Password policy management
- Active session monitoring
- User activity audit log
- Bulk user import and export

20.2.4 System Configuration

- Workflow configuration (approval chains, escalation rules)
- Notification templates (SMS, email, WhatsApp, push)
- Print template management (invoices, reports, labels, prescriptions)
- Numbering series configuration (UHID, invoice, order sequences)

- System-wide preferences (date format, currency, timezone, language)
- Feature flag management for phased rollout

21. Security Architecture and Compliance

21.1 Data Security

21.1.1 Encryption

- Data at rest: AES-256 encryption for database, file storage, and backups
- Data in transit: TLS 1.3 for all API communications
- Field-level encryption for PHI (Protected Health Information) fields
- Encryption key management with HSM (Hardware Security Module) support
- Database-level transparent data encryption (TDE)

21.1.2 Access Control

- Role-based access control (RBAC) with principle of least privilege
- Attribute-based access control (ABAC) for fine-grained policies
- IP whitelisting for administrative functions
- API key management with rate limiting
- OAuth 2.0 / OpenID Connect for third-party integrations
- JWT-based session management with short-lived tokens

21.1.3 Audit Logging

- Comprehensive audit trail for all data access and modifications
- Immutable audit logs (write-once, tamper-evident)
- Log retention policy: minimum 7 years for clinical data
- Real-time audit log streaming to SIEM systems
- User access reports for compliance audits
- Break-glass access logging with mandatory justification

21.2 Regulatory Compliance

Regulation	Scope	Key Requirements
HIPAA (US)	Patient health data	PHI protection, access controls, breach notification, BAA with vendors
GDPR (EU)	Personal data of EU residents	Consent management, right to erasure, data portability, DPO appointment
IT Act 2000 (India)	Electronic health records	Digital signature, data localization, reasonable security practices
DISHA (India - Draft)	Digital health data	Health data protection, consent framework, data fiduciary obligations
NABH	Hospital operations	Quality standards, patient safety, clinical audit, infrastructure compliance

NABL	Laboratory operations	Quality management, equipment calibration, proficiency testing, TAT compliance
FDA 21 CFR Part 11	Electronic records	Electronic signatures, audit trails, system validation

21.3 Disaster Recovery and Business Continuity

- Recovery Point Objective (RPO): less than 15 minutes for clinical data
- Recovery Time Objective (RTO): less than 1 hour for critical modules (ER, Pharmacy, Lab)
- Automated database backup: full daily, incremental every 15 minutes
- Cross-region replication for disaster recovery
- Failover testing: quarterly DR drills
- Offline mode for critical modules with sync-on-reconnect
- Data archival policy: active data (2 years), archive (5 years), cold storage (10+ years)

22. Deployment Architecture

22.1 Deployment Models

Model	Best For	Considerations
On-Premise	Large hospitals with IT teams, strict data sovereignty	Full control, higher upfront cost, in-house maintenance required
Cloud (Public)	Multi-site networks, rapid scaling, SaaS preference	Lower upfront, pay-as-you-go, vendor lock-in risk, data residency compliance
Hybrid	Hospitals needing data sovereignty with cloud analytics	Best of both, complex networking, requires VPN/private link
Private Cloud	Mid-large hospitals with compliance needs	Dedicated infrastructure, managed services, moderate cost

22.2 Infrastructure Requirements

22.2.1 Server Sizing Guidelines

Hospital Size	Concurrent Users	Application Servers	Database	Storage (Year 1)
Small (< 100 beds)	50-100	2 nodes (4 CPU, 16 GB RAM each)	Primary + 1 replica (8 CPU, 32 GB RAM)	500 GB (1 TB with PACS)
Medium (100-500 beds)	200-500	4 nodes (8 CPU, 32 GB RAM each)	Primary + 2 replicas (16 CPU, 64 GB RAM)	2 TB (10 TB with PACS)
Large (500+ beds)	500-2000	8+ nodes (16 CPU, 64 GB RAM each)	Clustered (32 CPU, 128 GB RAM per node)	10 TB+ (50 TB+ with PACS)
Multi-campus	2000+	Auto-scaling cluster	Distributed database cluster	Petabyte-scale with tiering

22.3 Network Requirements

- Internal network: minimum 1 Gbps LAN to all departments
- Redundant internet connectivity: primary + backup ISP
- VPN for remote access and inter-branch connectivity
- Network segmentation: clinical VLAN, administrative VLAN, guest VLAN, IoT/device VLAN
- Firewall with intrusion detection/prevention (IDS/IPS)
- Load balancer with SSL termination
- CDN for static assets and patient portal

22.4 Environment Strategy

Environment	Purpose	Data
Development	Feature development and unit testing	Synthetic/anonymized data
Staging / QA	Integration testing, UAT, performance testing	Anonymized copy of production
Pre-Production	Final validation, DR testing	Production-equivalent
Production	Live operations	Real patient and operational data
Training	User training and demonstrations	Sample data set

23. Comprehensive Data Model Overview

23.1 Core Entity Relationships

The following table summarizes the key entities across all modules and their primary relationships. This serves as a blueprint for database schema design, regardless of the chosen database technology (relational, document-oriented, or hybrid).

Domain	Core Entities	Key Relationships
Patient Master	Patient, PatientContact, PatientAddress, PatientDocument, PatientAllergy, NextOfKin, Consent	Patient is central; referenced by Visit, Admission, Invoice, LabOrder, etc.
Visits & Encounters	Visit, OPDVisit, Admission, EmergencyEncounter	Each linked to Patient; contains all clinical activity for an encounter
Clinical	ClinicalNote, Diagnosis, Prescription, PrescriptionItem, Referral, VitalSign	Linked to Visit/Admission; reference DrugMaster, ICD codes
Diagnostics	LabOrder, Sample, LabResult, RadiologyOrder, RadiologyReport	Linked to Visit; reference TestMaster/ExamMaster
Pharmacy	DrugMaster, DrugInteraction, PharmacyStock, Dispensing	DrugMaster referenced by Prescription, Stock, Dispensing
Billing	ChargeMaster, Tariff, Invoice, InvoiceLineItem, Payment, Advance, Refund, InsuranceClaim	Invoice links to Visit/Admission; LineItems link to services
Inventory	ItemMaster, Vendor, PurchaseOrder, GRN, StockEntry, StockTransfer	ItemMaster is central; referenced by department consumption
HR	Employee, Department, Designation, Attendance, Leave, Salary, PaySlip	Employee is central; linked to Attendance, Leave, Payroll
Blood Bank	Donor, BloodBag, BloodComponent, CrossMatch, BloodIssue, TransfusionReaction	Donor to BloodBag; Component to CrossMatch; Issue to Patient
Quality	QualityIndicator, IncidentReport, CAPA, AuditSchedule, InfectionSurveillance	Linked to departments, patients, and employees
System	User, Role, Permission, AuditLog, Configuration, Notification	User linked to Role; AuditLog tracks all entities

23.2 Data Governance

- Data classification: Public, Internal, Confidential, Restricted (PHI)
- Data retention policies per data category
- Data anonymization for analytics and research
- Master data management (MDM) for consistent reference data
- Data quality rules and validation at entry points
- Change data capture (CDC) for real-time data sync

24. API Design Guidelines

24.1 API Standards

All module APIs follow RESTful design principles with consistent patterns for resource naming, versioning, pagination, error handling, and authentication.

Aspect	Standard
Base URL	/api/v{version}/{module}/{resource}
Versioning	URL path versioning (v1, v2)
Authentication	Bearer token (JWT) in Authorization header
Content Type	application/json (default), application/fhir+json (FHIR endpoints)
Pagination	Cursor-based: ?cursor=xxx&limit=50 (max 100)
Filtering	Query parameters: ?status=active&department=cardiology
Sorting	?sort=created_at&order=desc
Error Format	{"error": {"code": "VALIDATION_ERROR", "message": "...", "details": [...]}}
Rate Limiting	X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset headers
CORS	Configurable allowed origins per environment
Idempotency	X-Idempotency-Key header for POST/PUT operations

24.2 API Module Summary

Module	Resource Prefix	Key Endpoints Count	FHIR Mapping
Patient	/api/v1/patients	~15	Patient, RelatedPerson
Appointments	/api/v1/appointments	~10	Appointment, Schedule, Slot
OPD	/api/v1/opd	~20	Encounter, Condition, MedicationRequest
IPD	/api/v1/ipd	~25	Encounter, Observation, CarePlan
Laboratory	/api/v1/lab	~20	ServiceRequest, DiagnosticReport, Observation
Radiology	/api/v1/radiology	~15	ImagingStudy, DiagnosticReport
Pharmacy	/api/v1/pharmacy	~15	Medication, MedicationDispense
Billing	/api/v1/billing	~20	Claim, Invoice, PaymentReconciliation
Inventory	/api/v1/inventory	~15	SupplyRequest, SupplyDelivery
HR	/api/v1/hr	~20	Practitioner, PractitionerRole
Blood Bank	/api/v1/bloodbank	~15	BiologicallyDerivedProduct

Admin	/api/v1/admin	~25	Organization, Location, HealthcareService
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24.3 Webhook / Event System

The HM-ERP publishes domain events for real-time integration and workflow automation:

Event	Payload Includes	Typical Consumers
patient.registered	UHID, demographics	External HIS, insurance portals
patient.admitted	Admission details, bed	Dietary, housekeeping, nursing
patient.discharged	Discharge summary	Follow-up system, patient portal
lab.result.validated	Test results, critical flag	EMR, notification service
pharmacy.dispensed	Prescription, drugs	Billing, MAR
billing.invoice.created	Invoice details	Accounting, insurance
inventory.stock.low	Item, current qty, reorder level	Purchase, notifications
incident.reported	Incident details	Quality team, management

25. User Interface and Experience Guidelines

25.1 Design Principles

- Clinical Efficiency: minimize clicks for common tasks (3-click rule for frequent actions)
- Context Preservation: patient context banner always visible during clinical workflows
- Error Prevention: confirmation dialogs for critical actions, undo support where possible
- Accessibility: WCAG 2.1 AA compliance, keyboard navigation, screen reader support
- Responsive Design: optimized for desktop (primary), tablet (nursing), and mobile (doctor rounds)
- Dark Mode: support for clinical environments (OT, radiology reading rooms)

25.2 Key Screen Descriptions

Screen	Primary Users	Key Elements
Patient Search/Registration	Front Desk	Search bar with filters, quick registration form, recent patients list
Doctor OPD Workstation	Doctors	Patient queue, clinical notes area, order panel, prescription pad, history timeline
Nursing Dashboard	Nurses	Patient list by bed, vital signs quick entry, MAR grid, care plan checklist
Bed Management Board	Nursing/Admin	Visual floor map, color-coded beds, occupancy stats, transfer drag-drop
Lab Worklist	Lab Techs	Sample list by section, result entry grid, validation queue, pending list
Pharmacy Counter	Pharmacists	Prescription queue, drug search, dispensing form, stock lookup
Billing Counter	Billing Staff	Patient charges summary, payment form, receipt printer, insurance panel
Admin Dashboard	Administrators	KPI widgets, alerts panel, quick actions, trend charts
Patient Portal	Patients	Appointment booking, reports download, bill payment, feedback form

25.3 Notification Strategy

Priority	Channel	Examples
Critical (P0)	Push notification + SMS + In-app alert + Sound	Critical lab values, emergency code, drug allergy override
High (P1)	Push notification + In-app alert	New admission, stat orders, blood request, OT schedule change
Medium (P2)	In-app notification	Lab results ready, appointment reminder, approval requests

Low (P3)	In-app notification (badge)	Report generated, stock alerts, system updates
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26. Implementation Roadmap

26.1 Phased Approach

The implementation is recommended in four phases, allowing the hospital to realize value progressively while managing change effectively.

Phase	Duration	Modules	Prerequisites
Phase 1: Foundation	3-4 months	Admin, Patient Registration, Appointment Scheduling, OPD, Billing (basic)	Infrastructure setup, master data migration, staff training (core)
Phase 2: Clinical Core	3-4 months	IPD, Bed Management, Nursing, Emergency, Pharmacy, Lab, Radiology	Phase 1 live, clinical staff training, instrument interfacing
Phase 3: Support Services	2-3 months	Inventory, HR/Payroll, Blood Bank, OT Management, Dietary	Phase 2 stable, department-specific data migration
Phase 4: Advanced	2-3 months	Insurance/TPA, Quality Management, Analytics/BI, Patient Portal, Teleconsultation	Phase 3 stable, historical data migration, executive training

26.2 Critical Success Factors

- Executive sponsorship from hospital leadership (CEO, CMO, CFO)
- Dedicated project team with clinical and IT representation
- Comprehensive change management and communication plan
- Phased go-live with parallel run period
- 24/7 support during go-live period (first 2 weeks per phase)
- Continuous training program with super-user model
- Data migration strategy with validation checkpoints
- Performance benchmarking and SLA monitoring from day one

26.3 Risk Mitigation

Risk	Impact	Mitigation
User resistance to change	High	Early stakeholder engagement, champion network, visible quick wins
Data migration errors	High	Automated validation, parallel run, rollback plan
Integration failures	Medium	API-first approach, integration testing environment, fallback procedures
Performance issues at scale	Medium	Load testing before go-live, horizontal scaling plan, caching strategy
Regulatory non-	High	Compliance review at each phase, external audit engagement

compliance		
Scope creep	Medium	Strict change control board, MoSCoW prioritization, phase gates

27. Glossary

Term	Definition
UHID	Unique Hospital Identification number assigned to each patient
OPD	Outpatient Department - where patients visit for consultation without admission
IPD	Inpatient Department - where patients are admitted for treatment
EMR/EHR	Electronic Medical/Health Record - digital version of patient charts
HL7 FHIR	Health Level 7 Fast Healthcare Interoperability Resources - healthcare data exchange standard
DICOM	Digital Imaging and Communications in Medicine - medical imaging standard
PACS	Picture Archiving and Communication System - medical image storage and retrieval
ICD-10/11	International Classification of Diseases - WHO standard for diagnosis coding
SNOMED CT	Systematized Nomenclature of Medicine - Clinical Terms for clinical documentation
LOINC	Logical Observation Identifiers Names and Codes - lab and clinical observation coding
RBAC	Role-Based Access Control - security model based on user roles
PHI	Protected Health Information - individually identifiable health data
TPA	Third Party Administrator - entity that processes insurance claims on behalf of insurers
MAR	Medication Administration Record - documents when medications are given to patients
TAT	Turnaround Time - time from order to result/completion
MLC	Medico-Legal Case - cases with legal implications requiring police intimation
NABH	National Accreditation Board for Hospitals and Healthcare Providers
NABL	National Accreditation Board for Testing and Calibration Laboratories
GRN	Goods Receipt Note - document recording receipt of goods from vendor
PACU	Post-Anesthesia Care Unit - recovery area after surgery
SBAR	Situation, Background, Assessment, Recommendation - clinical handoff framework
DXA	Document XML Architecture unit - 1/1440th of an inch (used in docx formatting)
EOQ	Economic Order Quantity - optimal order quantity minimizing total inventory costs
ABC Analysis	Classification of inventory items by consumption value (A=high, B=medium, C=low)
VED Analysis	Classification of items by criticality (Vital, Essential, Desirable)

END OF DOCUMENT

Hospital Management ERP - Comprehensive Documentation v1.0

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